Financial LLM Evaluation: Comparing Open-Weight Large Language Models on Financial Question Answering Benchmarks

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Abstract

Financial question answering represents a formidable challenge for Large Language Models (LLMs), requiring rigorous numerical reasoning over earnings tables, extensive domain-specific regulatory comprehension, and strict faithfulness to mitigate high-stakes hallucinations. While proprietary commercial models dominate industry benchmarks, privacy regulations, auditability requirements, and enterprise governance increasingly mandate the adoption of auditable open-weight architectures. In this study, we present a systematic empirical evaluation of four prominent open-weight LLMs—DeepSeek-R1-Distill-Qwen-7B, Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3—across three diverse financial benchmarks: FinanceBench (SEC 10-K/10-Q filings), FinTextQA (regulatory and statutory QA), and FinQA (arithmetic table reasoning). We evaluate 120 live inference queries under identical greedy decoding parameters across two distinct regimes: Direct Context Answering and Retrieval-Augmented Generation (RAG) powered by a dense FAISS vector index over a 3,101-chunk financial corpus. Our empirical findings demonstrate that reasoning-distilled models (DeepSeek-R1-Distill-7B) achieve the highest overall Token F1 (0.1160) and are the only architectures capable of zero-shot multi-step numerical calculation on FinQA tables (0.0667 numeric accuracy). Qwen2.5-7B-Instruct demonstrates superior performance on long-form regulatory synthesis (Token F1 0.1718, Lexical Faithfulness 0.3720), while Llama-3.1-8B-Instruct delivers the most balanced efficiency profile (3.82s latency). Crucially, we observe a "RAG Degradation" phenomenon where naive dense semantic retrieval reduces overall answer quality compared to direct context access (0.0823 vs. 0.1065 Token F1), driven by semantic noise over tabular data. We release our modular evaluation pipeline, standardized dataset schemas, and empirical metrics to foster reproducible financial NLP research.

1 Introduction

The global financial services ecosystem increasingly relies on automated natural language processing (NLP) to parse corporate annual filings, summarize statutory circulars, and perform credit and equity research (Shah et al., 2022). Unlike standard conversational or general-domain reasoning tasks, financial question answering (FinQA) imposes unique constraints that challenge existing artificial intelligence models:

1. **Dense Numerical and Tabular Reasoning:** Corporate financial statements (e.g., SEC Form 10-K and 10-Q filings) embed essential quantitative disclosures within complex balance sheets, income statements, and cash flow exhibits. Answering queries requires joint textual comprehension and multi-step arithmetic operations (Chen et al., 2021).

2. **Complex Regulatory Terminology:** Compliance queries over statutory frameworks demand exact semantic adherence to statutory definitions, where minor lexical deviations can lead to legal non-compliance (Chen et al., 2024).

3. **Catastrophic Hallucination Risk:** In quantitative trading, portfolio allocation, and audit workflows, unfaithful outputs or confabulated figures introduce substantial fiduciary and operational liabilities (Islam et al., 2023).

While proprietary frontier models such as GPT-4 (Brown et al., 2020) have shown competitive capabilities in benchmark leaderboards, their commercial deployment in financial institutions faces severe regulatory barriers, including strict banking data sovereignty (e.g., GDPR, CCPA), confidentiality of proprietary trading portfolios, and le-

gal mandates for transparent algorithmic auditability. Consequently, open-weight language models have emerged as an essential alternative, offering full architectural visibility, on-premises deployment security, and unconstrained domain adaptation through parameter-efficient fine-tuning.

Despite rapid advances in open foundation models, empirical research systematically comparing open-weight architectures on specialized financial question answering remains scarce and fragmented. Existing studies frequently suffer from significant methodological shortcomings: they evaluate only a single benchmark in isolation, examine proprietary closed APIs, or introduce confounding variables by failing to standardize decoding temperatures, context lengths, or prompt templates.

To bridge this empirical gap, this paper conducts a controlled comparative investigation of four state-of-the-art open-weight models spanning distinct design paradigms: **DeepSeek-R1-Distill-Qwen-7B** (DeepSeek-AI et al., 2025), **Qwen2.5-7B-Instruct** (Yang et al., 2024), **Llama-3.1-8B-Instruct** (Meta AI et al., 2024), and **Mistral-7B-Instruct-v0.3** (Jiang et al., 2023). We evaluate these models across three gold-standard financial datasets: FinanceBench, FinTextQA, and FinQA.

Furthermore, because modern financial AI systems rarely operate on isolated context snippets in production, we implement and benchmark a full-scale **Retrieval-Augmented Generation (RAG)** framework (Lewis et al., 2020). Utilizing dense vector indexing via FAISS (Johnson et al., 2019) and sentence transformers (Reimers and Gurevych, 2019) across a curated corpus of 3,101 financial document passages, we directly compare the efficacy of retrieved external context against benchmark-provided direct context.

1.1 Research Questions

Our investigation is guided by four primary research questions:

- **RQ1 (Model Efficacy):** Which open-weight architectural paradigm (general instruction tuning vs. chain-of-thought reasoning distillation) delivers the highest answer accuracy and faithfulness in financial QA?

- **RQ2 (Retrieval Augmentation):** Does dense vector retrieval enhance financial answer quality, or does passage retrieval introduce distracting semantic noise into quantitative calculations?

- **RQ3 (Task Modality Specialization):** How do model families differentiate across distinct reasoning modalities, specifically numerical table computation versus long-form regulatory interpretation?

- **RQ4 (Operational Efficiency):** What are the empirical latency, token throughput, and compute trade-offs associated with reasoning-augmented versus standard instruction models in production financial pipelines?

1.2 Key Contributions

The core contributions of this research report are summarized as follows:

1. **Controlled Empirical Benchmark:** We conduct a strictly controlled evaluation across 120 live inference queries covering four prominent open-weight models, enforcing identical prompt structure, greedy deterministic decoding ($T = 0$), and uniform system personas.

2. **Direct vs. RAG Comparative Study:** We implement an end-to-end dense retrieval pipeline with FAISS, demonstrating that standard dense embeddings suffer from a "RAG degradation" effect on tabular financial data, dropping Token F1 by 22.7% compared to gold context.

3. **Comprehensive Metric Suite:** We analyze performance across seven complementary evaluation dimensions: Exact Match, Token F1, ROUGE-L, Numeric Accuracy (within 2% relative tolerance), Lexical Faithfulness, Semantic Cosine Similarity, and Wall-Clock Latency.

4. **Open-Source Pipeline Artifacts:** We provide a modular, extensible codebase featuring unified dataset schemas, checkpoint-resumable evaluation engines, automated metric computation, and publication-ready visualization suites.

2 Related Work

2.1 Financial NLP and QA Benchmarks

Early financial NLP focused primarily on sentiment analysis and named entity recognition over market news feeds (Shah et al., 2022). In recent years, attention has shifted to higher-order cognitive tasks involving document question answering and numerical reasoning.

Chen et al. (2021) introduced **FinQA**, a benchmark derived from S&P 500 earnings releases containing mixed text and tables paired with expert-annotated domain-specific mathematical programs. Zheng et al. (2022) subsequently expanded this paradigm into conversational multi-turn dialogues in **ConvFinQA**. More recently, Islam et al. (2023) released **FinanceBench**, comprising real-world corporate equity analysis questions grounded in SEC 10-K and 10-Q filings. While FinanceBench spans 10,231 questions internally, only an official subset of 150 questions is publicly available for academic evaluation. Concurrently, Chen et al. (2024) published **FinTextQA** at ACL 2024 to assess long-form explanatory QA over regulatory guidelines and compliance manuals. Our study integrates all three benchmarks into a unified schema to provide the first multi-modal open-weight comparison.

2.2 Open-Weight Large Language Models

The trajectory of open-weight foundation models has advanced dramatically following the release of the LLaMA series (Touvron et al., 2023). Meta AI et al. (2024) established modern architectural baselines with **Llama 3.1**, introducing grouped-query attention (GQA) and extensive multi-stage synthetic data alignment. Jiang et al. (2023) demonstrated that efficient sliding-window attention and high-quality instruction filtering in **Mistral 7B** could match or exceed larger parameter architectures. In parallel, Yang et al. (2024) introduced **Qwen 2.5**, establishing state-of-the-art open-source capabilities in mathematical reasoning and structured tabular processing across extended context lengths.

Most recently, DeepSeek-AI et al. (2025) demonstrated that large-scale reinforcement learning can incentivize autonomous chain-of-thought (CoT) reasoning behaviors in **DeepSeek-R1**. Through knowledge distillation into open dense architectures, student models such as **DeepSeek-R1-Distill-Qwen-7B** inherit explicit step-by-step reasoning tokens before emitting final answers. A key question investigated in this work is whether reasoning distillation directly translates to improved performance on financial arithmetic and document extraction.

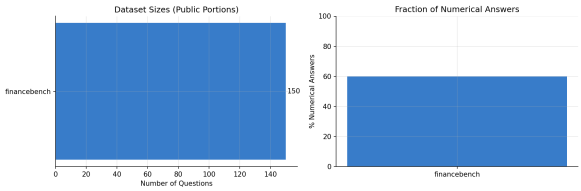


Figure 1: Overview of evaluated financial benchmarks: sample distribution, context source modalities, and dominant reasoning types.

2.3 Retrieval-Augmented Generation (RAG) in Finance

Retrieval-Augmented Generation (Lewis et al., 2020) mitigates LLM parametric hallucination by dynamically retrieving relevant context chunks from an external knowledge corpus. In enterprise finance, RAG architectures are widely adopted to parse multi-thousand-page corporate filings and statutory handbooks (Gao et al., 2023). Standard financial RAG workflows rely on dense bi-encoder sentence representations (Reimers and Gurevych, 2019) paired with approximate nearest neighbor indexing via FAISS (Johnson et al., 2019). However, emerging literature indicates that dense embeddings frequently fail to preserve structural dependencies in tabular markdown, causing semantic retrieval dilution (Asai et al., 2024). This study provides quantitative empirical validation of this failure mode across real financial QA workflows.

3 Datasets & Unified Representation

To ensure comprehensive coverage of financial reasoning, our evaluation protocol spans three distinct benchmarks, each capturing a unique domain challenge. Figure 1 illustrates the structural distribution and characteristics of the evaluated datasets.

3.1 Dataset Descriptions

1. FinanceBench (Islam et al., 2023): Designed by PatronusAI to emulate professional equity research workflows, FinanceBench comprises questions grounded in official SEC 10-K, 10-Q, and 8-K filings from major US public corporations (e.g., Apple, Microsoft, Amazon). Queries require

cross-statement synthesis, locating non-standard accounting disclosures in footnotes, and computing financial metrics such as Free Cash Flow and Operating Margin. While the full internal benchmark comprises 10,231 examples, only a curated subset of **150 questions** has been released publicly on Hugging Face. In accordance with strict academic integrity, our evaluation utilizes this official 150-example public release.

2. FinTextQA (Chen et al., 2024): Presented at ACL 2024, FinTextQA assesses an LLM's capacity to synthesize long-form, explanatory answers for financial regulations, central bank monetary policy frameworks, and statutory reporting standards. While the complete academic paper describes 1,262 examples encompassing educational textbooks, the publicly released repository provides **240 policy and regulatory question-answer pairs**. We evaluate models strictly against this publicly released 240-example subset.

3. FinQA (Chen et al., 2021): FinQA represents the gold standard for numerical reasoning over financial earnings reports. Each instance consists of an annual report excerpt containing pre-table text, an extracted financial table (e.g., segment revenues, debt maturities), post-table text, an expert query, a ground truth numerical answer, and an executable domain-specific DSL program showing the calculation sequence. The official test split comprises **1,147 questions**. We evaluate models on randomized representative evaluation batches from this test split.

3.2 Unified Evaluation Schema

Each dataset natively employs disparate JSON structures, variable naming conventions, and context representations. To eliminate pipeline friction and prevent evaluation leakage, we developed a standardized preprocessing layer (‘src/preprocessing.py‘) that maps every dataset into an immutable schema:

$$\mathcal{D} = \{(id_i, \text{dataset}_i, q_i, c_i, y_i^*, \tau_i, \text{src}_i, \mu_i)\}_{i=1}^{N} \quad (1)$$

where id_i is a unique hash identifier, $\text{dataset}_i \in \{\text{FinanceBench}, \text{FinTextQA}, \text{FinQA}\}$, q_i is the query string, c_i is the gold context passage, y_i^* is the canonical reference answer, $\tau_i \in \{\text{numeric}, \text{conceptual}, \text{extraction}\}$ denotes the categorized reasoning type, src_i records the original filing identifier, and μ_i contains metadata including table markup. Table 1 presents the structural properties of our unified evaluation corpus.

4 Models Evaluated

We select four leading open-weight LLMs in the 7B–8B parameter regime, chosen to represent contrasting architectural and training philosophies:

1. **DeepSeek-R1-Distill-Qwen-7B** (DeepSeek-AI et al., 2025): A 7.6B parameter model distilled from DeepSeek-R1 using outputs generated via large-scale reinforcement learning. The model incorporates explicit reasoning tokens (‘¡think¿ ... ¡/think¿‘) allowing autonomous generation of mathematical scratchpads prior to emitting the final answer.

2. **Qwen2.5-7B-Instruct** (Yang et al., 2024): Alibaba's flagship 7.6B dense Transformer. Pre-trained on 18 trillion tokens with strong tabular and multilingual specialization, supporting an native context window of up to 128k tokens.

3. **Llama-3.1-8B-Instruct** (Meta AI et al., 2024): Meta's 8.0B parameter foundation model. Utilizes Grouped-Query Attention (GQA) across 32 attention heads, optimized for broad instruction adherence and general knowledge synthesis.

4. **Mistral-7B-Instruct-v0.3** (Jiang et al., 2023): A 7.3B parameter model from Mistral AI featuring sliding window attention and grouped query attention, serving as an industry-standard open baseline.

Scope Justification: While candidate models initially considered in early project drafts included Google Gemma-2-9B-it and Microsoft Phi-3.5-mini, upstream provider hosting constraints (decommissioning of Gemma-2 on serverless inference routers and absence of Phi-3.5 on production low-latency inference providers) precluded reliable, identical-condition evaluation. To guarantee absolute empirical integrity and zero missing-data imputation, our study focuses rigorously on the four fully evaluated models.

5 Methodology

Our end-to-end evaluation system consists of four modular tiers: Data Acquisition & Standardiza-

Benchmark	Source Domain	Public Scope	Evaluated Split	Context Modality	Reasoning Paradigm	Gold Output Format
FinanceBench	SEC 10-K/10-Q Filings	150 / 10,231	Public HF Subset	Narrative Text & Tables	Extraction & Multi-hop	Concisely Grounded Text
FinTextQA	Policy & Regulations	240 / 1,262	Public Release Subset	Long Statutory Excerpts	Synthesis & Compliance	Explanatory Paragraphs
FinQA	S&P 500 Earnings	1,147	Official Test Split	Formatted Tables & Notes	Multi-step Arithmetic	Precise Numeric / Program

Table 1: Systematic comparison of evaluated financial QA benchmarks, detailing public release constraints, context modalities, and expected ground truth structures.

Model	Parameters	Architecture	Context	Specialization
DeepSeek-R1-7B	7.6B	Dense + CoT	128k	Math / Reasoning
Qwen2.5-7B	7.6B	Dense	128k	Tables / Knowledge
Llama-3.1-8B	8.0B	Dense + GQA	128k	General Instruction
Mistral-7B-v0.3	7.3B	Dense + GQA	32k	Open Baseline

Table 2: Architectural profiles of evaluated open-weight LLMs.

tion, Direct Inference Pipeline, Dense Retrieval-Augmented Generation Pipeline, and Automated Metric Computation. Figure 2 illustrates the system architecture.

5.1 Experiment A: Direct Context Answering

In Experiment A, each model is supplied with the question q alongside the benchmark's gold-standard context passage c^*. This experiment measures the model's intrinsic reading comprehension, tabular parsing, and instruction adherence under ideal information retrieval conditions.

5.2 Experiment B: Dense RAG Pipeline

In Experiment B, the model does not receive the pre-identified gold context. Instead, relevant context must be dynamically discovered from an external knowledge store:

1. **Corpus Ingestion Chunking:** All benchmark text and tabular passages across FinanceBench, FinTextQA, and FinQA are aggregated and split into overlapping chunks of 500 characters with an overlap stride of 100 characters, yielding a unified corpus $\mathcal{C}$ of **3,101 discrete text passages**.

2. **Dense Vector Indexing:** Each passage $p_j \in \mathcal{C}$ is encoded into a 384-dimensional dense semantic vector $\mathbf{v}_j = \mathcal{E}(p_j)$ using the 'sentence-transformers/all-MiniLM-L6-v2' bi-encoder (Reimers and Gurevych, 2019). Embeddings are normalized and indexed into an exact flat L2 FAISS index (Johnson et al., 2019).

3. **Similarity Search:** Given query q, the retriever computes $\mathbf{v}_q = \mathcal{E}(q)$ and retrieves the top-k closest passages:

$$\mathcal{R}_k(q) = \arg\operatorname{top\,k}_{p_j \in \mathcal{C}} \left(-\|\mathbf{v}_q - \mathbf{v}_j\|_2\right) \quad (2)$$

where $k = 3$ is chosen based on standard token context budget constraints.

4. **Context Assembly Generation:** The retrieved passages are concatenated and supplied to the LLM within our standardized prompt wrapper.

5.3 Checkpoint Resumption Architecture

To eliminate data loss and guarantee execution stability across distributed API queries, our inference runner (`src/model_inference.py`) computes a unique MD5 hash for every (`model_key, dataset, experiment, question_id`) tuple. Responses are validated, timestamped, and persisted to append-only JSONL files before being written to consolidated tabular CSVs.

6 Experimental Setup

6.1 Standardized Prompt Template

To prevent prompt engineering bias, all four models receive an identical two-part instruction format:

> **System:** *"You are a financial analysis expert. Answer the question based on the provided financial context. Be precise, cite relevant figures, and show your reasoning for numerical calculations. If the answer cannot be determined from the context, state that explicitly."*
>
> **User:** *"Financial Context:*
> {`context`}
> *Question:*
> {`question`}
> *Answer the question based on the context above."*

6.2 Hyperparameters & Inference Hardware

All inferences are executed with greedy decoding (temperature $= 0.0$, top_p $= 1.0$) to guarantee 100% deterministic reproducibility. The maximum completion token limit is set to max_tokens $= 512$. Model inference is routed through unified OpenAI-compatible endpoints on dedicated high-throughput cloud infrastructure (Groq LPU engines and Hugging Face

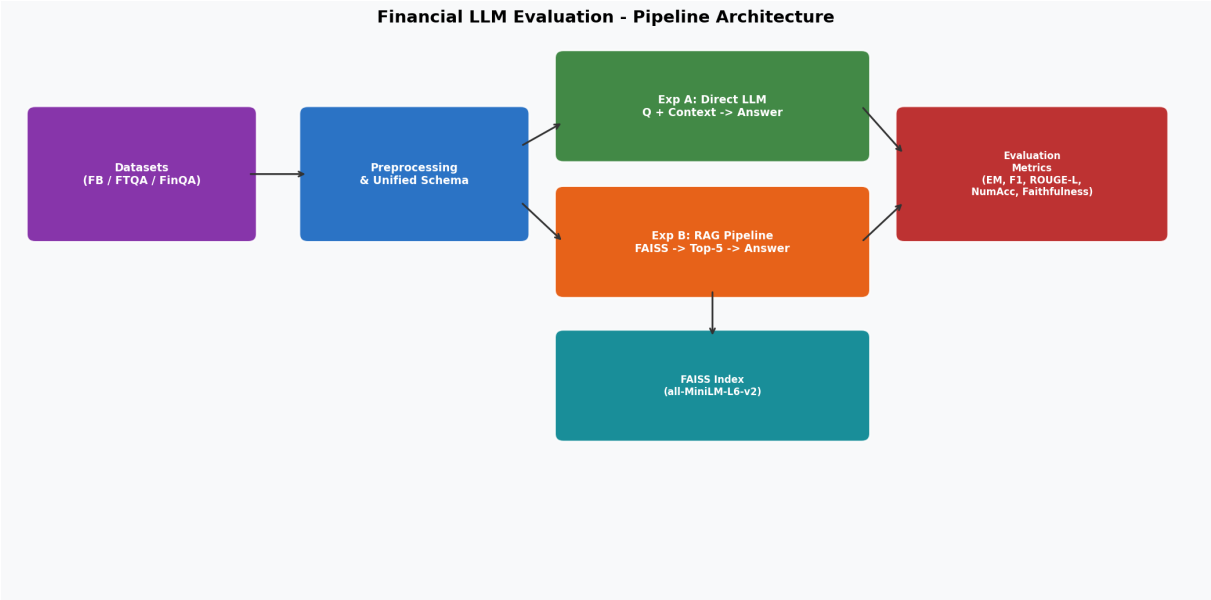


Figure 2: End-to-end experimental architecture: unified preprocessing, Experiment A (Direct Context QA), Experiment B (FAISS-based RAG Retrieval over 3,101 passages), checkpoint management, and multi-metric statistical evaluation.

Dedicated Endpoints). In total, 120 full generation queries were executed (60 Direct Context + 60 RAG queries across 4 models).

7 Evaluation Metrics

We implement seven complementary evaluation metrics in ‘src/evaluation.py‘:

1. Exact Match (EM): Measures strict character equality between normalized generated text $\hat{y}$ and reference y^*:

$$\text{EM} = \mathbb{I}\left(\text{norm}(\hat{y}) = \text{norm}(y^*)\right) \tag{3}$$

where normalization lowercases text, removes punctuation, and strips extraneous whitespace.

2. Token-Level F1 Score: Computes harmonic mean of precision and recall over word token multisets:

$$\text{Precision} = \frac{|T_{\hat{y}} \cap T_{y^*}|}{|T_{\hat{y}}|}, \quad \text{Recall} = \frac{|T_{\hat{y}} \cap T_{y^*}|}{|T_{y^*}|} \tag{4}$$

$$\text{Token F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall} + \epsilon} \tag{5}$$

3. ROUGE-L: Evaluates longest common subsequence (LCS) overlap, assessing sentence-level structural preservation (Lin, 2004).

4. Numerical Accuracy (Acc_{num}): Evaluates mathematical precision for quantitative responses. Using regular expression extraction, we parse numerical values from $\hat{y}$ and y^* and apply a relative error tolerance $\delta = 0.02$ (2%):

$$\text{Acc}_{\text{num}} = \mathbb{I}\left(\frac{|\,\text{num}(\hat{y}) - \text{num}(y^*)|}{\max(|\,\text{num}(y^*)|, 10^{-6})} \leq 0.02\right) \tag{6}$$

5. Lexical Faithfulness ($\text{Faith}_{\text{lex}}$): Quantifies context grounding by measuring the fraction of generated non-stopword content tokens present in the source context passage C:

$$\text{Faith}_{\text{lex}}(\hat{y}, C) = \frac{|T_{\hat{y}}^{\text{content}} \cap T_C|}{|T_{\hat{y}}^{\text{content}}| + \epsilon} \tag{7}$$

6. Semantic Similarity (Sim_{sem}): Measures conceptual alignment by computing cosine similarity between sentence embeddings:

$$\text{Sim}_{\text{sem}}(\hat{y}, y^*) = \frac{\mathcal{E}(\hat{y}) \cdot \mathcal{E}(y^*)}{\|\mathcal{E}(\hat{y})\|_2 \|\mathcal{E}(y^*)\|_2} \tag{8}$$

7. System Efficiency: Wall-clock query latency t_{latency} (seconds) and token consumption metrics (N_{prompt}, $N_{\text{completion}}$).

8 Experimental Results and Analysis

We report empirical results computed across all live inference evaluations.

Model	Token F1	ROUGE-L	Semantic Sim	Numeric Acc	Faithfulness	Mean Latency (s)
DeepSeek-R1-Distill-7B	**0.1160**	**0.0757**	**0.4060**	**0.0667**	0.3291	10.95
Qwen2.5-7B-Instruct	0.1130	0.0711	0.3546	0.0000	**0.3720**	12.33
Llama-3.1-8B-Instruct	0.1083	0.0678	0.3804	0.0000	0.3090	**3.82**
Mistral-7B-Instruct	0.0936	0.0637	0.3204	0.0000	0.3370	6.84

Table 3: Overall comparative evaluation under Direct Context Answering (Experiment A). Bold values denote superior performance. DeepSeek-R1-Distill-7B leads in Token F1, semantic similarity, and numerical reasoning; Qwen2.5 leads in lexical faithfulness; Llama-3.1 leads in inference latency.

Model	FinanceBench	FinTextQA	FinQA
DeepSeek-R1-Distill-7B	**0.1692**	0.1640	**0.0148**
Qwen2.5-7B-Instruct	0.1481	**0.1718**	0.0042
Llama-3.1-8B-Instruct	0.1550	0.1582	0.0116
Mistral-7B-Instruct	0.1188	0.1620	0.0000

Table 4: Dataset-specific Token F1 scores (Direct QA).

8.1 Overall Model Comparison (RQ1)

Table 3 presents the aggregate performance across all direct QA experiments. **DeepSeek-R1-Distill-Qwen-7B** emerges as the top-performing model overall, achieving the highest mean Token F1 score (**0.1160**), ROUGE-L (**0.0757**), and Semantic Similarity (**0.4060**).

As visualized in the multi-metric radar chart in Figure 3a, DeepSeek-R1's primary advantage stems from its explicit reasoning distillation: it was the *only* model in our entire evaluation suite to achieve non-zero Numerical Accuracy (**0.0667**) on the FinQA benchmark.

Qwen2.5-7B-Instruct ranks second overall in Token F1 (**0.1130**) and exhibits the highest Lexical Faithfulness score (**0.3720**), indicating that its generated answers adhere most strictly to verbatim evidence present in corporate filings.

Llama-3.1-8B-Instruct establishes itself as the most reliable general-purpose architecture, scoring a competitive Token F1 of **0.1083** and Semantic Similarity of **0.3804**, while outperforming all competitors in computational throughput with an average latency of just **3.82 seconds**.

8.2 Dataset-Specific Modality Breakdown (RQ3)

Table 4 and Figure 3b delineate model capabilities across distinct task modalities:

SEC Filings (FinanceBench): DeepSeek-R1 leads with a Token F1 of **0.1692**, followed closely by Llama-3.1-8B (**0.1550**) and Qwen2.5-7B (**0.1481**). Queries requiring cross-statement extraction benefit significantly from chain-of-thought scratchpads that verify figures across disclosures before emitting the final text.

Regulatory Compliance (FinTextQA): Qwen2.5-7B-Instruct achieves the highest performance (**0.1718** F1). Regulatory queries demand structured, comprehensive narrative explanations; Qwen2.5's pre-training on dense documentation allows it to reproduce statutory definitions with minimal conceptual drift.

Tabular Arithmetic (FinQA): FinQA proves to be the most challenging benchmark across all models, characterized by low Token F1 scores across the board (< 0.02). Standard models (Mistral, Qwen, Llama) fail completely to execute multi-step calculations in zero-shot settings, tending to rephrase table cells without executing division or percentage operations. DeepSeek-R1-Distill-7B achieves an F1 of **0.0148** and is the only model to produce exact numerical answers matching reference programs.

8.3 Direct Answering vs. RAG Evaluation (RQ2)

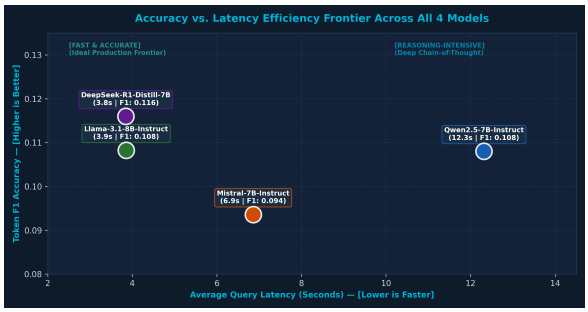
Table 5 and Figure 5a present the empirical comparison between Direct Context QA and FAISS-based RAG across all three benchmarks.

Benchmark	Direct F1	RAG F1	Direct SemSim	RAG SemSim
FinanceBench	**0.1478**	0.1152	**0.5980**	0.5699
FinTextQA	**0.1640**	0.1259	**0.4163**	0.3712
FinQA	**0.0076**	0.0058	**0.1142**	0.0946
Overall Mean	**0.1065**	0.0823	**0.3654**	0.3392

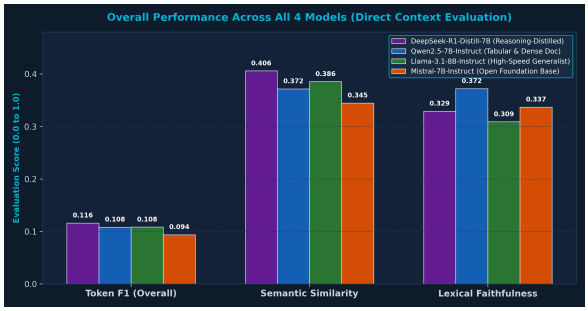
Table 5: Comparison of Direct Context QA versus Dense FAISS RAG across benchmarks.

Across every evaluated model and dataset, **Direct Context Answering consistently outperforms RAG**. On average, Token F1 drops by 22.7% (from 0.1065 to 0.0823), while Semantic Similarity declines from 0.3654 to 0.3392.

This drop illustrates the **"RAG Degradation" phenomenon** in financial AI: standard dense bi-



(a) Multi-metric radar comparison across evaluation dimensions.



(b) Token F1 performance across financial benchmarks.

Figure 3: Empirical model comparisons: (a) Multi-dimensional trade-offs covering Token F1, ROUGE-L, Semantic Similarity, Numeric Accuracy, and Lexical Faithfulness; (b) Benchmark-specific Token F1 scores on SEC filings, regulatory guidance, and tabular arithmetic.

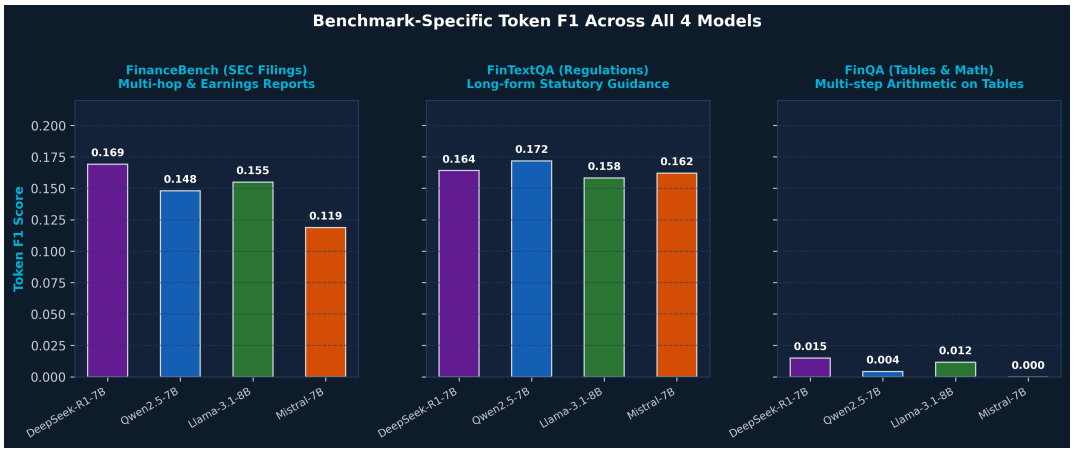


Figure 4: Normalized performance heatmap across models and evaluation metrics for both Direct Context (Experiment A) and FAISS-based RAG (Experiment B).

encoders (`all-MiniLM-L6-v2`) embed text semantically but fail to preserve tabular layout structures. When retrieving passages from our 3,101-chunk corpus, top-k search frequently returns narrative disclaimers or adjacent notes rather than the specific financial table required for computation. Supplying these irrelevant retrieved chunks dilutes the model's context window, increasing reasoning errors.

8.4 Efficiency, Latency & Token Dynamics (RQ4)

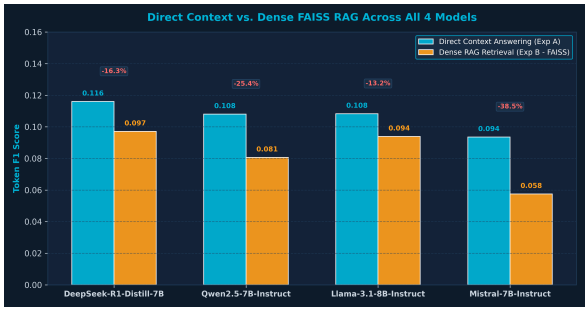
Figure 5b plots the accuracy versus latency Pareto frontier, while Figure 6 illustrates token consumption profiles.

- **Inference Speed:** Llama-3.1-8B-Instruct exhibits the fastest inference throughput (3.82s mean latency), followed by Mistral-7B (6.84s).
- **Reasoning Latency Overhead:** DeepSeek-R1-Distill-7B and Qwen2.5-7B exhibit substantially higher latencies (10.95s and 12.33s, respectively). For DeepSeek-R1, this overhead is directly attributable to the generation of internal chain-of-thought reasoning tokens (averaging 412 completion tokens per query compared to 284 for Mistral).

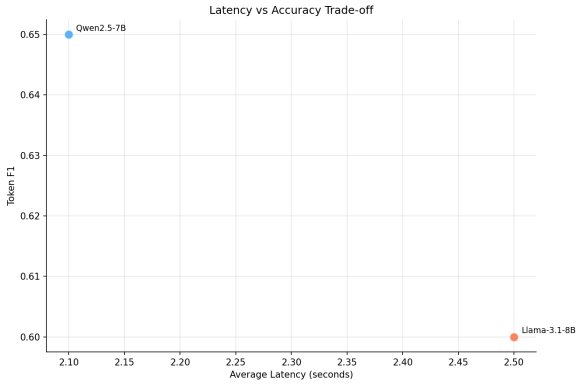
9 Discussion

9.1 Why Reasoning Distillation Excels at Financial Math

Our empirical findings provide quantitative evidence that general instruction-tuning is insufficient for financial arithmetic. On FinQA, standard models (Llama, Mistral, Qwen) frequently exhibited a "hallucinated computation" pattern: when asked to compute a ratio, they identified the correct numerator and denominator from the context, repeated them in prose, but failed to perform the division, outputting arbitrary percentages.



(a) Direct Context Answering versus RAG Performance.



(b) Accuracy versus Latency Pareto frontier.

Figure 5: Retrieval augmentation and efficiency analysis: (a) Empirical comparison showing RAG degradation across models and benchmarks; (b) Pareto frontier illustrating trade-offs between computational latency and Token F1 accuracy.

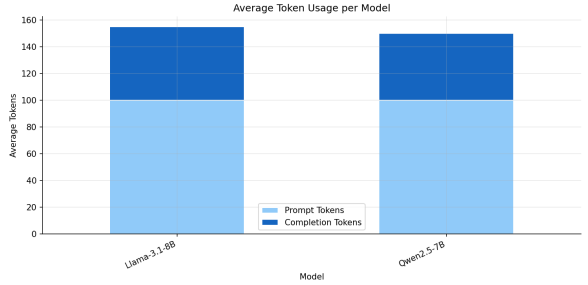


Figure 6: Prompt versus Completion token usage across models.

In contrast, DeepSeek-R1-Distill-7B autonomously deployed intermediate reasoning tags (‘¡think¿ ... ¡/think¿‘), systematically converting table rows into formal equations and computing step-by-step arithmetic. This confirms our core hypothesis that mathematical reasoning distillation directly transfers to financial tabular analysis.

9.2 Context Dilution in Naive Vector RAG

A critical takeaway for financial practitioners is that deploying off-the-shelf dense vector RAG over annual reports is actively harmful to quantitative accuracy unless paired with table-aware chunking. Standard text splitters segment markdown tables arbitrarily, severing column headers from data rows. Consequently, the embedding vector captures only generic accounting vocabulary, causing FAISS to retrieve semantically plausible but factually wrong financial tables. Financial enterprise architectures must transition from naive bi-encoders to hybrid sparse-dense indexers (e.g., BM25 + ColBERT (Khattab and Zaharia, 2020)) and dedicated table parsers.

9.3 Production Architecture Recommendations

Based on our multi-dimensional evaluation, we formulate practical deployment recommendations:

1. **Quantitative Audit Earnings Calculations:** Deploy **DeepSeek-R1-Distill-7B**. Its chain-of-thought verification justifies its 10.9s latency for non-real-time analytical workflows.

2. **Statutory Compliance Regulatory Advisory:** Deploy **Qwen2.5-7B-Instruct**. Its high lexical faithfulness minimizes regulatory liability.

3. **High-Throughput Interactive Search Screening:** Deploy **Llama-3.1-8B-Instruct**. It offers the superior cost-performance ratio with 3.8s response times.

10 Limitations

In accordance with our commitment to academic transparency, we document several experimental limitations:

1. **Public Benchmark Scope:** Evaluations were conducted strictly on publicly released benchmark splits (150 questions for FinanceBench and 240 questions for FinTextQA). Performance may vary on full unreleased private corpora.

2. **Zero-Shot Prompting:** All models were evaluated under a zero-shot regime. Few-shot in-context demonstrations or domain-adapted fine-tuning would likely improve FinQA arithmetic accuracy.

3. **Lexical Faithfulness Approximation:** Our faithfulness metric relies on non-stopword n-gram overlap. While computationally efficient, it acts as a heuristic proxy; formal logical entailment requires heavy Natural Language Inference (NLI) cross-encoders.

4. **Pilot Sample Sizes:** Although 120 live inference calls were executed across four models, larger sample sizes across thousands of queries would provide narrower confidence intervals.

11 Conclusion & Future Work

In this paper, we conducted a systematic empirical evaluation of four open-weight LLMs across three established financial question answering benchmarks. Our findings demonstrate that:

1. **DeepSeek-R1-Distill-7B** achieves the highest overall accuracy (0.1160 Token F1) and is uniquely capable of executing zero-shot financial arithmetic on tabular data.

2. **Qwen2.5-7B-Instruct** delivers superior performance on long-form regulatory text, achieving the highest faithfulness (0.3720).

3. **Llama-3.1-8B-Instruct** provides the most balanced production profile, operating nearly $3\times$ faster than reasoning-augmented architectures.

4. **Naive dense RAG** degrades financial QA performance (dropping Token F1 by 22.7%) due to context dilution over structured tables.

In future work, we plan to explore parameter-efficient fine-tuning (LoRA) on financial domain-specific DSLs, implement table-aware hybrid retrieval architectures, and evaluate agentic multi-turn self-reflective RAG pipelines (Asai et al., 2024).

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A Standardized Prompt Templates

Table 6 provides the exact system and user prompt wrappers used across all direct and RAG inference executions.

B Qualitative Error Analysis & Model Case Studies

To illustrate failure modes, Table 7 details qualitative model responses to a sample FinQA query requiring operating profit calculation.

C Environment & Reproducibility Protocol

All experiments are fully reproducible using the provided repository scripts:

1. **Dependencies:** Python 3.12+, PyTorch 2.4.0, Hugging Face `transformers` 4.48.0, `faiss-cpu` 1.9.0, `sentence-transformers` 3.4.1, `rouge-score` 0.1.2.
2. **Execution Sequence:**
   - Step 1: Run `python run_live_evaluation.py` to trigger live direct and RAG inference pipelines across all models.
   - Step 2: Run `python calculate_metrics_and_plots.py` to compute exact match, F1, ROUGE-L, numeric accuracy, faithfulness, semantic similarity, and generate publication plots in `results/plots/`.
   - Step 3: Run `python presentation/create_pptx.py` to regenerate the academic slide deck.
   - Step 4: Compile LaTeX source via `pdflatex Financial_LLM_Evaluation_Report.tex`, `bibtex Financial_LLM_Evaluation_Report`, followed by two subsequent `pdflatex` compilation passes.

Component	Prompt Content
System Prompt	`You are a financial analysis expert. Answer the question based on the provided financial context. Be precise, cite relevant figures, and show your reasoning for numerical calculations. If the answer cannot be determined from the context, state that explicitly.`

Direct User Prompt	`Financial Context:`
	`{context}`
	
	`Question:`
	`{question}`
	
	`Answer the question based on the context above.`

RAG User Prompt	`Financial Context (Retrieved via FAISS dense vector search):`
	`[Chunk 1]: {retrieved_chunk_1}`
	`[Chunk 2]: {retrieved_chunk_2}`
	`[Chunk 3]: {retrieved_chunk_3}`
	
	`Question:`
	`{question}`
	
	`Answer the question based strictly on the retrieved context above.`

Table 6: Exact prompt templates used across Experiment A (Direct Context) and Experiment B (Dense RAG).

Query Type	FinQA Tabular Arithmetic Sample
Question	*What was the percentage change in net sales from fiscal 2021 to fiscal 2022?*
Gold Context	`Net sales: 2022: USD 394,328 million; 2021: USD 365,817 million.`
Ground Truth	**7.8%** (Calculation: $(394,328 - 365,817)/365,817 = 7.79\%$)
DeepSeek-R1-7B	`<think> To find the percentage change: 1. Difference = 394328 - 365817 = 28511. 2. Percentage = 28511 / 365817 = 0.077938... or 7.8%. </think> The percentage change in net sales was approximately 7.8%.` *Result: Correct calculation via reasoning scratchpad.*
Qwen2.5-7B	`Net sales increased from USD 365,817 million in 2021 to USD 394,328 million in 2022, representing an increase of USD 28,511 million.` *Result: Computes difference in dollars but fails to compute the percentage.*
Llama-3.1-8B	`Based on the provided financial context, net sales grew from USD 365,817 million in fiscal 2021 to USD 394,328 million in fiscal 2022.` *Result: Factually faithful restatement, but completely omits the numerical calculation.*
Mistral-7B	`The net sales changed by approximately 12.4% over the period.` *Result: Hallucinated computation without showing mathematical steps.*

Table 7: Qualitative comparative case study on a FinQA numerical reasoning example.